

RadGPT: A System Based on a Large Language Model That Generates Sets of Patient-Centered Materials to Explain Radiology Report Information

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Abstract

Purpose: The 21st Century Cures Act final rule requires that patients have real-time access to their radiology reports, which contain technical language. The objective of this study was to use a novel system called RadGPT, which integrates concept extraction and a large language model (LLM), to help patients understand their radiology reports.

Methods: RadGPT generated 150 concept explanations and 390 question-and-answer pairs from 30 radiology report impressions from between 2012 and 2020. The extracted concepts were used to create concept-based explanations, as well as concept-based question-and-answer pairs for which questions were generated using either a fixed template or an LLM. Additionally, report-based question-and-answer pairs were generated directly from the impression using an LLM without concept extraction. One board-certified radiologist and four radiology residents rated the material quality using a standardized rubric.

Results: Concept-based LLM-generated questions were of significantly higher quality than concept-based template-generated questions ($P < .001$). Excluding those template-based question-and-answer pairs from further analysis, nearly all (>95%) of RadGPT-generated materials were rated highly, with at least 50% receiving the highest possible ranking from all five raters. No answers or explanations were rated as likely to affect the safety or effectiveness of patient care. Report-level LLM-based questions and answers were rated particularly highly, with 92% of report-level LLM-based questions and 61% of the corresponding report-level answers receiving the highest rating from all raters.

Conclusions: The educational tool RadGPT generated high-quality explanations and question-and-answer pairs that were personalized for each radiology report, unlikely to produce harmful explanations, and likely to enhance patient understanding of radiology information.

Key Words: RadGPT, concept extraction, large language model, radiology reports, patient safety

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INTRODUCTION

Starting on April 5, 2021, the US 21st Century Cures Act final rule mandates immediate electronic availability of health record information to patients. Electronic health record information includes radiology reports, which contain medical jargon not easily understandable to a layperson. On one hand, immediate access to health information promotes greater transparency and allows patients to be more involved in and informed about their medical care. Studies suggest that most people prefer instant access to their own health records [1], including radiology reports [2]. However, access to such information before explanation by a health care professional can lead to misinterpretation, anxiety, and possible harm [1,3]. This early access also leads to more

patient phone calls to ordering providers, thereby consuming limited time and contributing to physician burnout [4].

Although strategies have been proposed to modify radiology reports in response to the Cures Act [5-7], radiology reporting has remained largely the same: a means for physician-to-physician communication rather than an educational tool geared toward patients. The average US adult reads at an eighth grade reading level [8], which is the highest level recommended for health materials by the Centers for Disease Control and Prevention, the National Institutes of Health, and CMS [9]. However, only 4% of radiology reports are readable at the eighth grade level [8]. Therefore, there is a critical need to demystify radiology reports. As it is not feasible to provide real-time explanations from a radiologist or another trained medical professional, automated methods are needed, and large language models (LLMs) may be the solution. LLMs are deep learning-based generative artificial intelligence models pre-trained on vast amounts of text data to process and produce human language. LLMs can be used for many natural language processing tasks, including language generation, summarization, translation, and classification. ChatGPT (OpenAI) is an LLM-based interactive chatbot that boasts more than 180 million users at the time of this report. LLMs such as ChatGPT have already shown incredible promise in the field of medicine [10,11], including acquiring enough medical knowledge to pass the United States Medical Licensing Examination [12,13], facilitating social media-based pharmacovigilance to combat the opioid epidemic [14], providing diabetes self-management tips [15], and answering questions related to breast cancer screening [16]. A particular strength of ChatGPT is its “human touch”: one study demonstrated that ChatGPT’s responses to patient questions on a public social media forum were not only high quality but also empathetic [17].

The ability of LLMs to provide both accurate and patient-centered information suggests that LLMs can play a crucial role in radiology reporting [18-20]. Initial studies indicate that ChatGPT can effectively simplify radiology reports into plain language [21-24]. We aim to harness the capabilities of LLMs to generate a clinical tool that can educate patients about their radiology reports by explaining the concepts that each individual patient considers relevant. Using GPT-4 (OpenAI’s latest iteration of GPT) and the RadGraph concept extraction tool [25], we created a framework of explanations and question-and-answer pairs to explain impressions of radiology reports to patients and answer follow-up questions that patients may have. We assess the quality of the materials generated by this framework for implementation as a patient-friendly clinical tool.

METHODS

Data Collection

Approval for this work was obtained from the Stanford University institutional review board, and the need to obtain written informed consent was waived because of the retrospective nature of the study. A sample of 30 radiology reports was obtained from a retrospective database of radiology reports from between 2012 and 2020 at our academic medical center from each of the following study types: (1) abdominal and pelvic CT, (2) head CT, (3) sinus CT, (4) neck MR, (5) spine MR, and (6) chest radiography. For each type of study, reports were processed using the Universal Sentence Encoder [26]. To obtain this sample of 30 reports, we selected 5 reports from each of the six types that exhibited the lowest cosine similarity to guarantee semantic diversity, resulting in a total of 30 reports. Reports were deidentified and postprocessed by removing proper nouns using “hide in plain sight” rule-based methods [27].

Concept Extraction and Generation of Explanations and Question-and-Answer Pairs

To assess the efficacy of our RadGPT tool, several methods of question-and-answer generation were developed. First, using the RadGraph algorithm, we extracted all key concepts (anatomy and observation) and concept relations (modify, located at, suggestive of) from each radiology impression. We postprocessed the semantic graph output by RadGraph by concatenating annotations to form longer entities. For example, the observations “opacification” and “infection” with the relation “suggestive of” would form the concept “opacification suggestive of infection.” For each report, we randomly chose 5 concepts from all extracted ones, ensuring that there were no duplicates whenever possible. A total of 150 concepts were generated from the 30 radiology reports, with 144 of these being unique (Supplemental Table 1).

Our RadGPT tool incorporated a single LLM, which was OpenAI’s most recent iteration of GPT, GPT-4. For each concept, a single explanation was generated using this LLM (“concept-level explanation”), using the following prompt: “Explain in 3 sentences or less the concept ‘[CONCEPT]’ at a 5th grade reading level given the following context: ‘[IMPRESSION].’ Do not discuss symptoms or treatment.” Additionally, for each concept, two questions were generated, one that was entirely LLM produced (“concept-based LLM-generated question”) and one that was fixed on the basis of a template (“concept-based template-based question”). The concept-based LLM-generated questions were created using the prompt “Generate exactly 1 question a patient would ask about this medical explanation: ‘[CONCEPT EXPLANATION].’ Do not ask about symptoms or treatment.” The

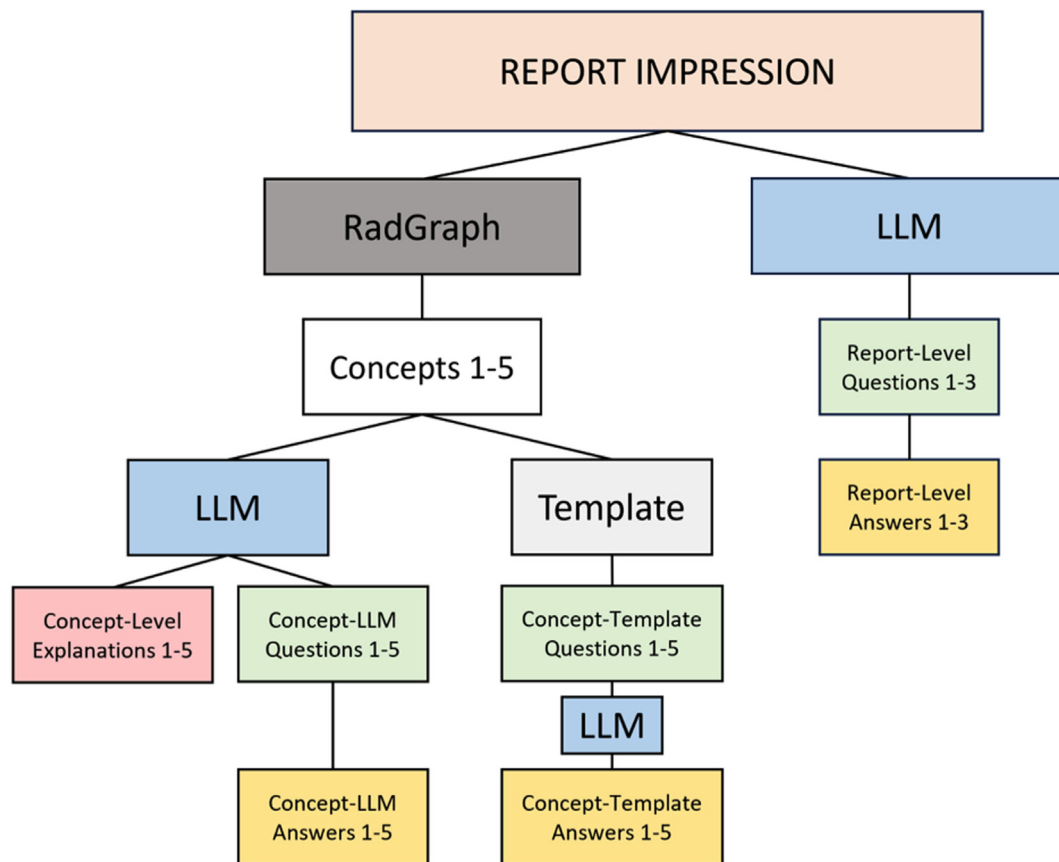


Fig. 1. Schematic of content generation for each report impression. RadGraph was used to extract five concepts from each report impression. For each concept, a concept-based explanation was generated using RadGPT. For each concept, two questions were also generated, one generated by the RadGPT large language model (LLM) and one using a template-based structure. Answers were generated by RadGPT for each of these two questions. Additionally, three report-level questions were generated with the RadGPT LLM, with corresponding RadGPT-generated answers. Using this approach, 13 question-and-answer pairs (3 report-level question-and-answer pairs and 10 concept-level question-and-answer pairs) and 5 concept-level explanations were generated for each report impression.

concept-based template-based questions were generated by choosing one of the three question structures that was most relevant to the concept: (1) what is <concept>, (2) what is the function of <concept> (if the concept is a normal anatomic structure), and (3) what is the cause of <concept> (if the concept is an abnormal structure or finding). For both LLM-generated and template-based question types, the LLM was used to generate answers using the prompt “Answer in 3 sentences or less the question ‘[QUESTION]’ at a 5th grade reading level given the following context: ‘[IMPRESSION].’ Do not discuss symptoms or treatment.” Finally, we prompted the LLM to generate directly from the impression three questions and their corresponding answers (“report-level LLM-generated questions and answers”), without using RadGraph to create predefined concepts. Using this approach, a total of 5 concept-level explanations and 13 question-and-answer pairs (10 concept-level pairs and 3 report-level pairs) were generated per impression (Fig. 1).

For all questions and answers, the LLM was asked to generate language at the fifth grade reading level, a level that the majority (80%) of Americans can understand and the maximum reading level at which The Joint Commission recommends that patient education materials be written [28]. Each unique type of explanation, question, or answer was assessed for Flesch-Kincaid grade reading level using Microsoft 365 Word version 2308. The group of report-level LLM-generated questions had an average Flesch-Kincaid grade reading level of 9.4, which likely reflected these questions’ asking about complex terminology used in the original report impressions. All of the other materials had an average Flesch-Kincaid grade reading level less than 8.0, which was our goal to meet the criteria of the Centers for Disease Control and Prevention, the National Institutes of Health, and CMS. RadGPT was prompted to limit explanations and answers to three sentences each to help remove irrelevant information and promote conciseness of responses.

Table 1. Rubric used to rate an explanation or answer (on a five-point scale) or a question (on a three-point scale)

Likert Rating	Interpretation of Rating for Answers or Explanations	Interpretation of Rating for Questions
1.0	Information is likely to very negatively affect the safety or effectiveness of the patient's future medical care	Definitely not relevant
2.0	Information could negatively affect the safety or effectiveness of the patient's future medical care	Possibly relevant or definitely relevant but phrased in an unnatural way
3.0	Information would negatively affect patient understanding of the patient's current medical condition but likely would not influence future medical care	Definitely relevant
4.0	Information provided would not improve patient understanding of the patient's imaging findings or medical condition, or the information provided would improve the patient's understanding but is phrased in an unnatural way	
5.0	Information provided is accurate and would improve patient understanding of the patient's imaging findings or medical condition, and the information is phrased in a natural way	

Expert Ratings

Five radiology-trained physicians (four radiology residents [S.E.H., P.S., A.J., and C.O.] in postgraduate years 3-5 of residency training and one attending radiologist [C.P.L.]) served as raters for the study. To promote interrater consistency, raters convened after reviewing explanations and question-and-answer pairs for 5 report impressions (not included in the final set of 30 reports) to create a detailed written scoring rubric for assessing the quality of explanations, questions and answers. A three-point rating scale was used for questions to evaluate the relevance of each question, and a five-point Likert-type rating scale was used for answers and explanations to evaluate the potential impact of the provided information on both patient understanding and patient medical care (Table 1). Importantly, answers and explanations with ratings of 1 or 2 were judged as likely or having the potential to pose harm to the patient's medical care, whereas those with ratings of 3 or higher were judged as not posing harm to the patient's medical care. After consensus was achieved and discrepancies were addressed, the finalized rubric was applied by individual raters to the explanations and question-and-answer pairs for the 30 reports. All of the raters participated in the selection of the templates to be used for the concept-based template-generated questions, but during the rating process, the raters were not told which concept-based questions were template generated or LLM generated.

Statistical Analysis

GraphPad Prism, Google Sheets, and Python were used to perform statistical analyses. Because of the multireader,

multicase study design, to account for the intra- and inter-rater variabilities from reports, concepts, and raters, mixed-effects models were performed to compare the ratings of concept-based questions that were either template based or LLM generated and their corresponding answers. A two-sided P value of $<.05$ was considered to indicate statistical significance. To assess interrater agreement for each type of RadGPT-generated material, the percentage agreement among the raters and Fleiss's κ coefficients were calculated. No adjustments were made for the multiplicity of concepts.

RESULTS

There was high interrater agreement for all types of RadGPT-generated material, with a Fleiss's κ coefficient of 0.66 and percentage complete rater agreement of 51% across all answer and explanation ratings and a Fleiss's κ coefficient of 0.65 and percentage complete rater agreement of 71% across all question ratings, indicating substantial agreement [29]. To aid patients first learning about key concepts of interest in the radiology report, customized concept-level explanations were generated as described in the "Methods" section. The average rating of all of the 150 RadGPT-generated concept-level explanations was 4.8 out of 5.0 (Fig. 2, Table 2). Importantly, 95% of concept explanations had an average rating of 4 or higher. Furthermore, 50% of all concept explanations were given the highest possible rating of 5.0 by every rater. In contrast, 5% of concept-level explanations had an average rating less than 4.0. Only 1% of explanations were rated as 2 by any single reviewer, implying the explanation was considered potentially detrimental to a patient's safety or

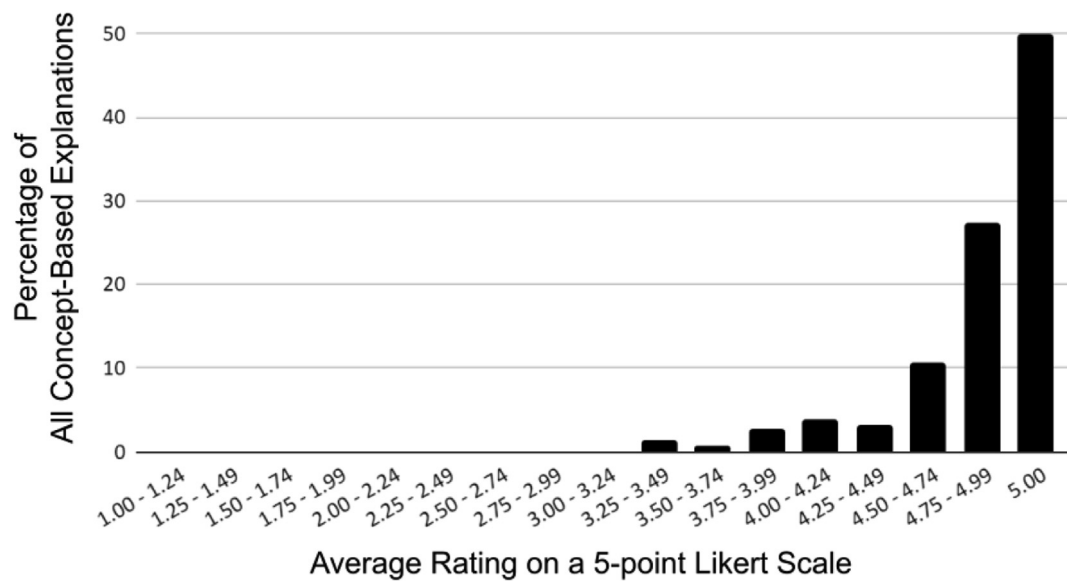


Fig. 2. Histogram of ratings of 150 concept-level explanations. The x axis is the range of average ratings of the explanations on a five-point Likert-type scale, and the y axis is the percentage of the explanations with average scores within the indicated range.

future medical care ([Supplemental Figure 1](#); examples are provided in [Supplemental Table 2](#)). No explanations were rated as 1 by any rater.

Both template-based and LLM-generated questions about the concept of interest were generated using RadGPT. The 150 LLM-generated questions were rated significantly higher in quality than the 150 template-based questions (average rating of 2.9 out of 3.0 vs 2.6 out of 3.0, $P < .001$ from mixed-effects model; [Supplemental Figure 2](#)). Forty-seven percent of template-based questions and 77% of LLM-generated questions received the highest possible rating from all raters (examples are provided in [Table 3](#)). Thirteen percent of template-based questions and no LLM-generated questions had an average rating less than 2.0 (examples are provided in [Supplemental Table 3](#)).

The quality of the answers to these concept-based questions was assessed in a similar manner. The quality of the 150 answers to LLM-generated questions was rated significantly higher than the 150 answers to template-based questions, although the absolute difference was small (average rating of 4.7 out of 5 vs 4.6 out of 5, $P = .001$ from mixed-effects model; [Fig. 3](#)). Forty-two percent of answers to template-based questions and 52% of answers to LLM-generated questions received the highest rating of 5.0 from all raters (examples are provided in [Table 2](#)). In contrast, 8% of answers to template-based questions and 5% of answers to LLM-generated questions had average ratings less than 4.0. Five percent of answers to template-based questions and 2% of answers to LLM-generated

questions were rated as 2 by any single reviewer, implying that the answers were considered potentially detrimental to a patient's safety or future medical care ([Supplemental Figure 3](#); examples are provided in [Supplemental Tables 4](#) and [5](#)). No answers received a rating of 1 by any rater.

In the final phase of the experiment, RadGPT was used to generate three pairs of questions and answers tailored to each individual radiology report but not limited to a pre-designated concept. The overall average rating of the 90 report-level LLM-generated questions was 3.0 out of 3.0 ([Supplemental Figure 4](#)). Ninety-two percent of report-level LLM-generated questions received the highest rating of 3.0 from all raters. No report-level LLM-based question had an average rating less than 2.0, with the lowest average rating being 2.6 out of 3.0. The overall average rating of the corresponding report-level answers to these questions was 4.7 out of 5.0 ([Fig. 4](#)). In total, 61% of report-level answers received the highest rating of 5.0 from all raters. Only 4% of report-level answers had an average rating less than 4.0. Three percent of answers were rated as 2 by any single reviewer, implying the explanation was considered potentially detrimental to a patient's safety or future medical care ([Supplemental Figure 5](#); examples are provided in [Supplemental Table 6](#)). No answers were rated as 1 by any rater. Therefore, the majority of both the questions and answers generated by RadGPT regarding a specific radiology report were considered high-quality and given the highest possible rating by all raters.

Table 2. Summary of the ratings of each category of content

Content	Quantity of Content Rated	Average Rating	Percentage of Content With Maximum Average Rating	Percentage of Content With Low Rating
Concept-level explanations	150	4.8 out of 5.0	50% (5.0 from every rater)	1% (rated 2 by any rater)
Concept-level LLM-generated questions	150	2.9 out of 3.0	77% (3.0 from every rater)	0% (average rating below 2.0)
Concept-level template-based questions	150	2.6 out of 3.0	47% (3.0 from every rater)	13% (average rating below 2.0)
Answers to concept-level LLM-generated questions	150	4.7 out of 5.0	52% (5.0 from every rater)	2% (rated 2 by any rater)
Answers to Concept-Level Template-Based Questions	150	4.6 out of 5.0	42% (5.0 from every rater)	5% (rated 2 by any rater)
Report-level LLM-generated questions	90	3.0 out of 3.0	92% (3.0 from every rater)	0% (average rating below 2.0)
Answers to report-level LLM-generated questions	90	4.7 out of 5.0	61% (5.0 from every rater)	3% (rated 2 by any rater)

Note: Raters ranked each question on a three-point Likert-type scale and each answer on a five-point Likert-type scale. No explanations or answers received a rating of 1 by any reviewer. LLM = large language model.

DISCUSSION

Since the passage of the 21st Century Cures Act, patients can first access their radiologic health information directly from their electronic health records rather than through discussion with their health care teams. This direct access improves patient autonomy and informed medical decision making. However, radiologic health information contains complex medical concepts and technical terminology that can trigger patient anxiety and promote the use of unreliable online medical information, possibly leading patients to pursue unnecessary or harmful medical care. Therefore, there is a critical need for validated, accurate, and patient-centered information that is tailored to an individual radiology report, written at an appropriate reading level and available in real time. We have created and optimized RadGPT, a LLM-based patient-centered education tool for radiology reports that identifies specific concepts discussed in the radiology report and offers patient education in 3 formats: (1) LLM-generated explanations of these specific concepts; (2) concept-specific questions, either template based or LLM generated in structure, with corresponding LLM-generated answers; and (3) LLM-generated question-and-answer pairs that are tailored to a specific radiology report but not limited to a particular concept. Although high-quality report-level materials were generated by the LLM alone without the use of RadGraph concept extraction, the automated extraction of concepts from individual radiology reports by RadGPT allows the generation of material focused on a particular concept within an individual radiology report without human prompting or curation. This concept-based material is a one key advantage of RadGPT compared with an LLM model alone because it allows the rapid generation of personalized content that is not just summarizing or discussing a radiology report as a whole but permits a patient to focus on a specific concept in the report that is of greatest interest or concern.

RadGPT offers tailored, patient-centered educational material, about both the overall radiology report and specific concepts within the report, that poses a low risk to patient safety and in the vast majority of cases is ready for immediate use without further human curation. The only RadGPT content that was produced with human input were concept-based, template-based questions because these questions were based on a set of human-generated templates. These template-based questions and their corresponding answers were found to be significantly lower in quality than the corresponding concept-based questions generated by an LLM. Our radiology-trained physician raters found that all three forms of non-template-based RadGPT content were high quality, and an average rating between the highest and second highest possible ratings was

Table 3. Examples of concept-based LLM-generated questions and corresponding answers that were given the highest possible ratings by every rater

Concept	LLM-Based Question	Average Question Rating	Answer Provided by RadGPT	Average Answer Rating
Possible carcinoid	What tests will be done to confirm whether the mass is a carcinoid, lymphoma, or something else?	3.0	The doctors will do more tests like scans or maybe even take a small piece of the mass to look at under a microscope. These tests will help them figure out if the mass is a carcinoid, lymphoma, or something else. They need to know what it is so they can decide the best way to help.	5.0
Nephrolithiasis	What could have caused these kidney stones to form in my kidneys?	3.0	Kidney stones can form when your body has too much of certain types of substances, like calcium or uric acid, which can build up and harden into stones. Not drinking enough water can also cause kidney stones because it allows these substances to gather and stick together. Certain foods, like too much salt or sugar, can also increase the chance of getting kidney stones.	5.0
Lacunar	What causes the tiny blood vessels in the brain to get blocked, leading to lacunar infarcts?	3.0	Lacunar infarcts, or small strokes, happen when tiny blood vessels in the brain get blocked. This can be caused by things like high blood pressure or diabetes. It's like when a road gets blocked and cars can't get through, but in this case, it's blood that can't get to parts of the brain.	5.0
Residual chronic collection	Will this "residual chronic collection" of old blood in my brain dissolve over time or does it need to be removed?	3.0	The "residual chronic collection" of old blood in your brain might slowly go away over time. But, it's also possible that doctors might need to remove it. It really depends on your specific situation.	5.0

Note: Raters ranked each question on a three-point Likert-type scale and each answer on a five-point Likert-type scale. LLM = large language model.

assigned to 95% of all concept-level explanations, 100% of all concept-based LLM-based questions and 95% of the corresponding answers, and 100% of report-level LLM-based questions and 96% of the corresponding answers. Furthermore, all five raters gave the highest possible score to 50% of all concept-level explanations, 77% of all concept-based LLM-based questions and 52% of the corresponding answers, and 92% of report-level LLM-based questions and 61% of the corresponding answers. No explanations were rated as likely to have a negative effect on patient

safety. Therefore, RadGPT content appears poised to fulfill the critical need for real-time, accurate, personalized, and patient-centered educational materials, which would ultimately allow patients to access explanations through hyperlinks to any or all of the concepts mentioned in their reports. These tools address the persistent patient-centered need for focused, personalized explanations of the specific concepts within radiology reports that individual patients find most important or concerning, in addition to a general simplification of the entire radiology report.

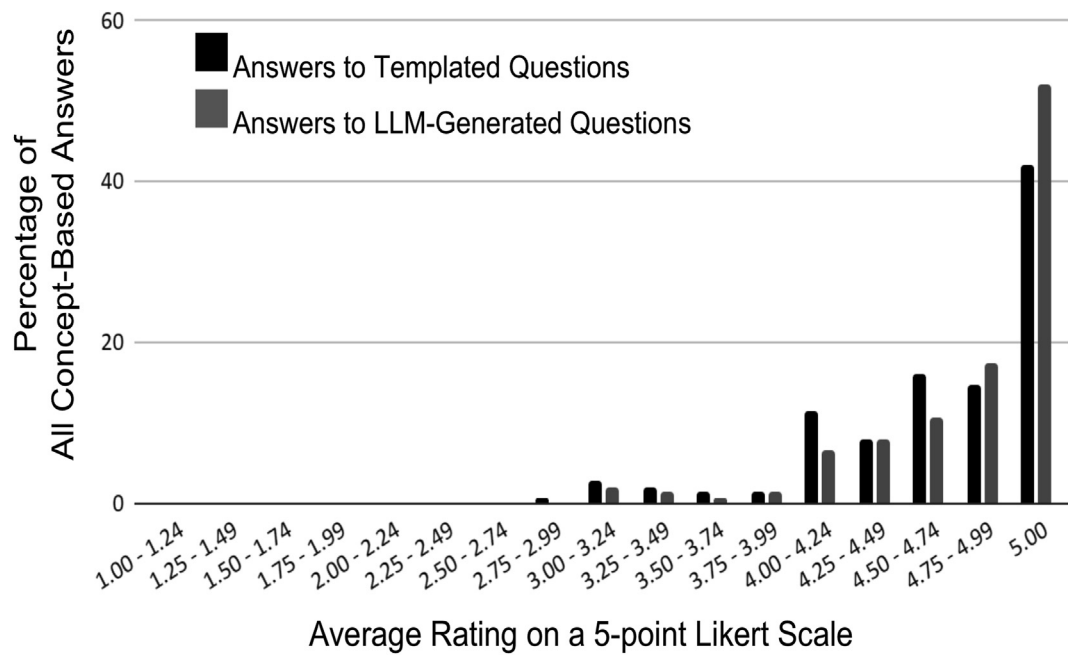


Fig. 3. Histogram of ratings of 150 answers to concept-based, template-based questions and 150 answers to concept-based large language model (LLM)-generated questions. The x axis is the range of average ratings of the answers on a five-point Likert-type scale, and the y axis is the percentage of the answers with average scores within the indicated range.

There were limitations to this study. First, the assessments of the explanations, questions, and answers in this study were performed solely by physicians, and the justification for each rating was not collected. Ultimately, future

assessments by patients, including those with a wide range of health literacy, will be necessary to confirm that these explanations are appropriate for the targeted patient audience. The goal of this work is to validate the safety of these

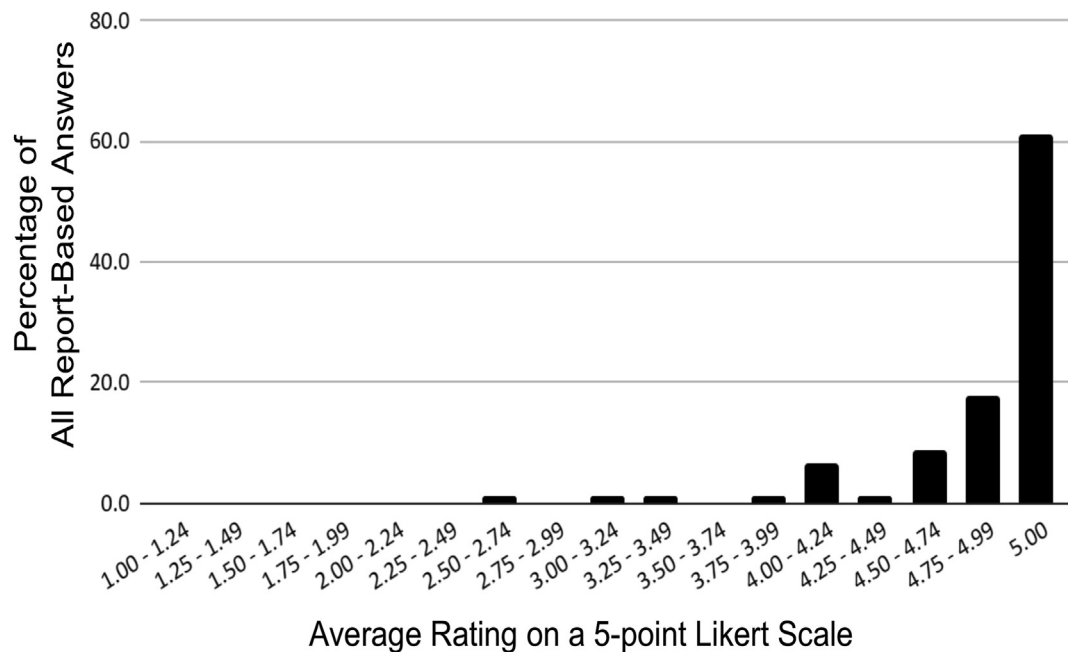


Fig. 4. Histogram of ratings of 90 answers to report-level large language model-based questions. The x axis is the range of average ratings of the answers on a five-point Likert-type scale, and the y axis is the percentage of the answers with average scores within the indicated range.

materials before they are offered to patients, especially because the most valuable future patient assessment would be of materials explaining each patient's own unique radiology report.

Second, although we assessed radiology reports discussing a wide range of radiology modalities and anatomic structures, the size of this study did not allow the inclusion of examples of every type of radiology examination or subgroup analyses of individual types of radiology examinations; assessment of RadGPT performance with additional types of radiology examinations would be necessary before RadGPT is used clinically for those reports.

Third, because the reports were deidentified before further analysis, potential problems related to the processing of proper nouns and other personalized information were not evaluated.

Finally, the RadGPT tool evaluated in the present study incorporated only a single concept extraction tool and a single LLM, GPT-4, and therefore did not assess or compare the performance of other concept extraction tools or LLMs, which would presumably have different overall performance in this setting.

In summary, we present RadGPT, a patient education tool offering real-time, high-quality customized content to improve patients' understanding of their own radiology reports. RadGPT creates consistently high-quality LLM-generated explanations and question-and-answer pairs that are tailored to individual radiology reports. These results support RadGPT as a safe tool with the potential to effectively enhance patient engagement and health outcomes without increasing the burden of health care workers and to promote health equity regardless of patients' education levels or medical literacy.

TAKE-HOME POINTS

- The novel system RadGPT, which integrates a concept extraction tool and an LLM, created individualized, patient-centered explanations and question-and-answer pairs for 30 radiology reports spanning multiple modalities and anatomic regions.
- Average ratings between the highest and second highest possible ratings were assigned to 95% of concept-level explanations, 95% of answers to concept-based LLM-based questions, and 96% of answers to report-level LLM-based questions.
- No materials were rated as likely to negatively affect patient safety.

ADDITIONAL RESOURCES

Additional resources can be found online at: <https://doi.org/10.1016/j.jacr.2025.06.013>.

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